

Tooth Development

Ectomesenchymal origin

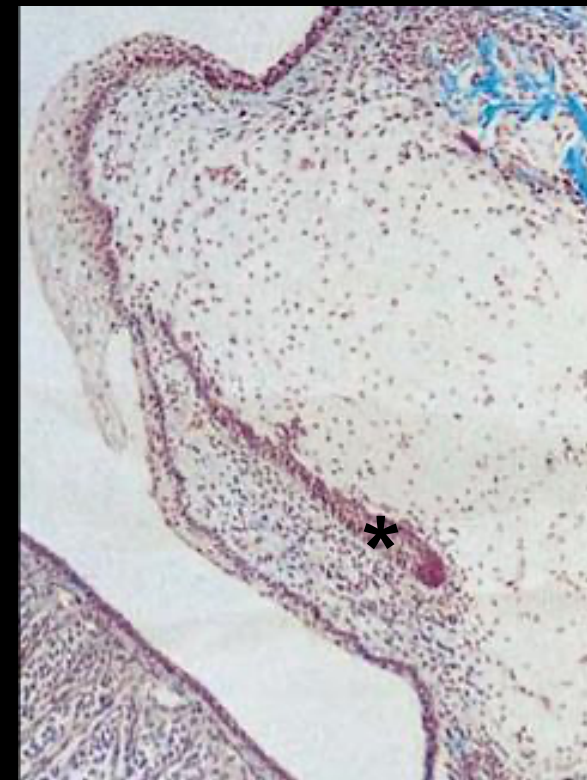
Tooth formation arise from **epithelial-mesenchymal interaction** in the **6th week of intrauterine life**

The development is between **oral epithelium** & **mesenchyme** derived from neural crest cells

Firstly the basal layer of the epithelium of the oral cavity forms a C-shaped structure in the upper & lower jaws which is called as **dental lamina**



Epithelial band at the
6th week of intra-uterine life



Dental lamina

Dental lamina

gives

Dental buds
(10 in each jaw)

The **1st histological sign of tooth development** is the appearance of **mesenchymal tissue** & **capillary networks** beneath the dental epithelium of the primitive oral cavity

Tooth Development

Ectomesenchymal origin

At about **7th week** the **primary epithelial band** divides into:

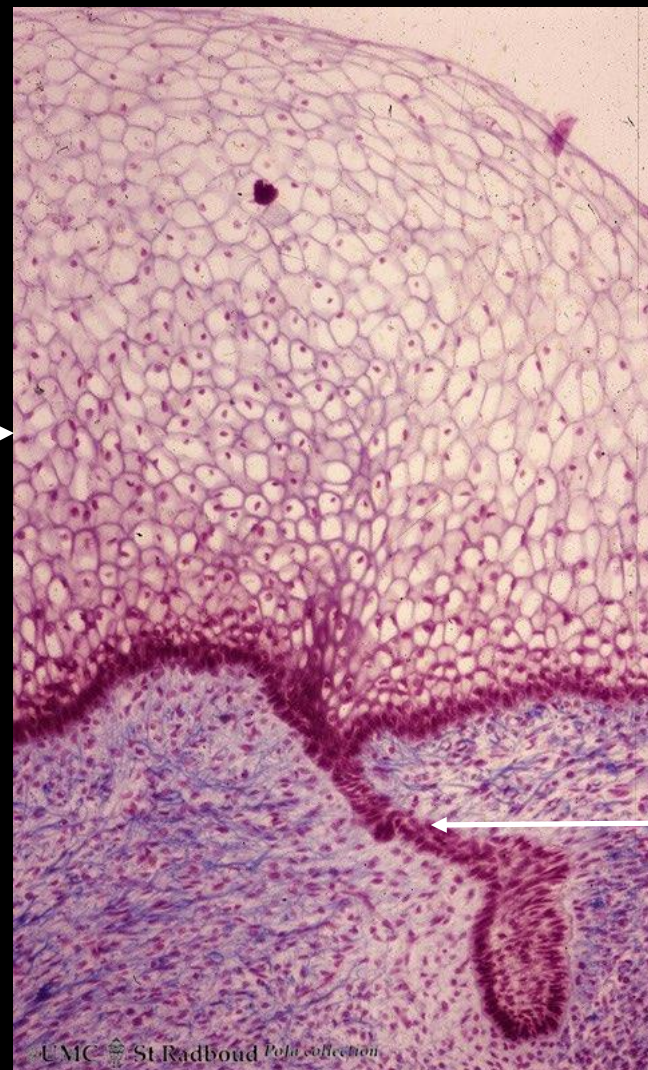
A **lingually located process** (dental lamina) → Contributes in the development of the teeth

A **buccally located process** (vestibular lamina) → Contributes in the formation of the vestibule of the mouth

All deciduous teeth arise from **dental lamina**

The **primitive oral cavity (stomodeum)**
is lined by stratified squamous epithelium
called the **oral ectoderm**

Most of the connective tissue cells
underlying the oral ectoderm are of
ectomesenchyme in origin



The tooth development begins in
the anterior portion of the jaw &
proceeds posteriorly

Regulation of tooth formation from
incisors to molars is generated by
HOX genes in the mesenchyme

Dental lamina

Tooth Development

Phases of tooth development

1) **Initiation phase**

Sites of the future teeth are established
Dental lamina is formed

2) **Morphogenesis phase**

Shape of the tooth is determined

3) **Histogenesis phase**

Differentiation of cells forms dental tissues
Mineralized tissues → enamel, dentine, cementum
Unmineralized tissues → dental pulp, periodontium

Stages of tooth development

1) **Bud stage**

2) **Cap stage**

3) **Bell stage**

Signals for development involves growth factors:

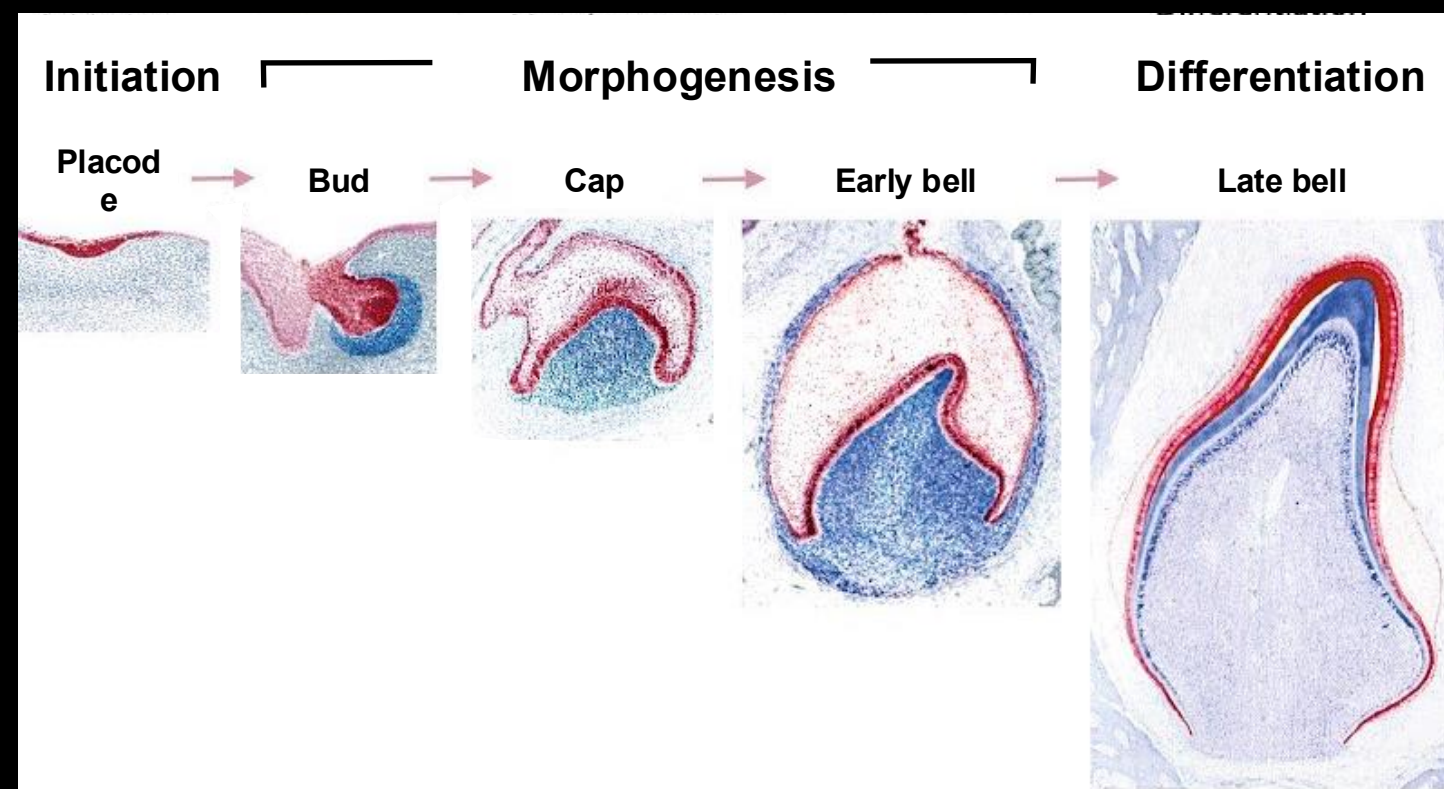
- 1) **BMP** (bone morphogenetic proteins)
- 2) **Wnt**
- 3) **FGF** (fibroblast growth factor)

Secreted factor:

Shh (sonic hedgehog)

Transcription factor:

MSX 1 & 2



Tooth Development

BUD STAGE

The **enamel organ (A)** appears as a spherical / ovoid
epithelial condensation that is **poorly**
morphodifferentiated & histodifferentiated

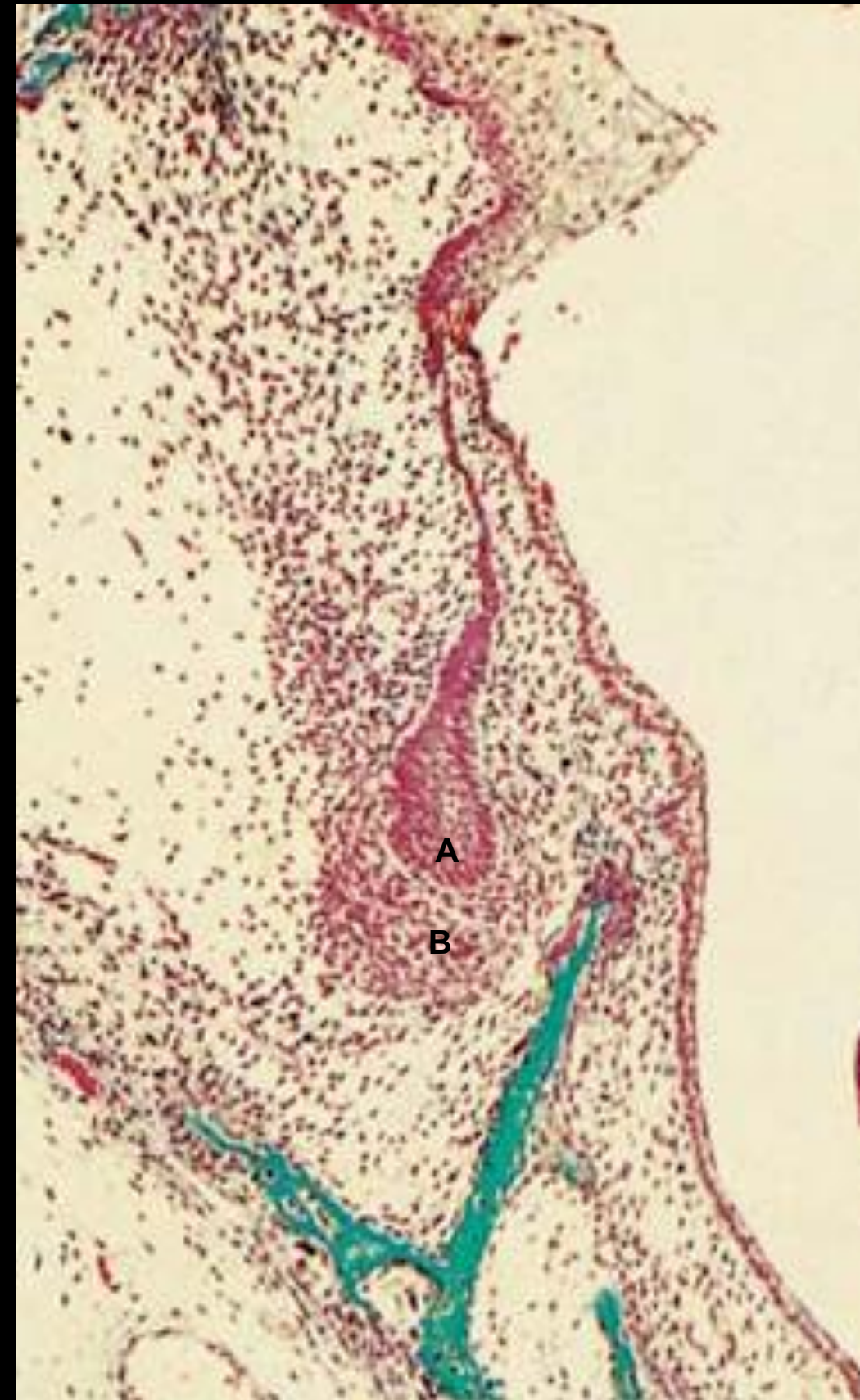
It is surrounded by **mesenchyme (B)**

The cells of the tooth bud have:

Higher RNA content than oral epithelium

Lower glycogen content

increased oxidative enzyme activity



Tooth Development

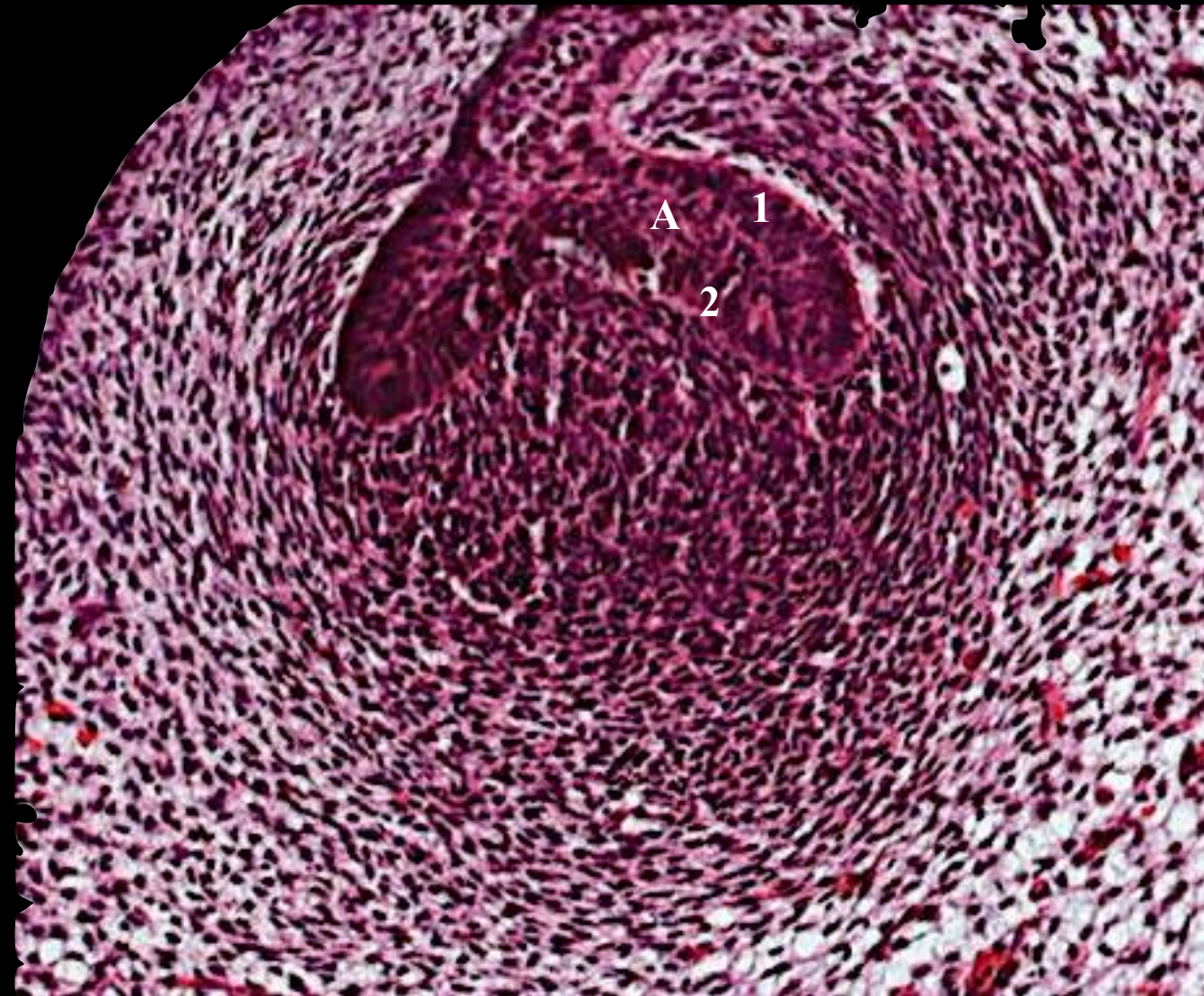
CAP STAGE (early stage)

By 11th week → Morphogenesis progresses

Deeper surface of the enamel organ (A) invaginate to form a *cap-shaped* structure

The cap consists of :

- 1) Outer enamel epithelium
- 2) Inner enamel epithelium



Tooth Development

CAP STAGE (late stage)

About **12th week** → **Central cells of the enlarging enamel organ** become separated (but they maintain in contact by desmosomes)

Between the inner & outer enamel epithelium high amount of **glycosaminoglycans** are found which result in formation of the **stellate reticulum**

The cells of the external enamel epithelium remain **cuboidal**

The cells of the internal enamel epithelium become more **columnar**



Mesenchymal cells continue to proliferate beneath the **internal enamel epithelium** to form **dental papilla**.

A = stellate reticulum

B = external enamel epithelium

C = internal enamel epithelium

D = dental papilla

E = dental follicle

Tooth Development

Dental cap grows & the indentation deepens → the tooth takes on the appearance of a **bell (bell stage)**

Mesenchyme cells of the **papilla** near to the inner enamel epithelium differentiate into **odontoblasts**

Odontoblasts → **Dentin**

By thickening of the dentin layer, odontoblasts leave a thin cytoplasmic process (**dental process**) behind in the dentin

The odontoblast layer stays throughout the life of the tooth & continuously provides **predentin**

The remaining cells of the **dental papilla** form the **pulp** of the tooth.

The **inner enamel epithelium** differentiate into **ameloblasts/adamantoblasts (enamel formers)**

Ameloblasts → **Enamel prisms**

A group cells in the inner enamel epithelium forms the **enamel knot** that *regulates early tooth development*

*When the enamel thickens, the ameloblasts retreat into the stellate reticulum. Here they regress, temporarily leaving a thin membrane (**dental cuticle**) on the surface of the enamel. After eruption of the tooth, this membrane gradually sloughs off !!*

Tooth Development

EARLY BELL STAGE

By the **14th week**, morphodifferentiation & histodifferentiation of the tooth germ lead to the early **bell stage**

The configuration of the internal enamel epithelium broadly maps out the occlusal pattern of the crown of the tooth

LATE BELL STAGE

The late bell stage (appositional stage) of tooth development is associated with the formation of the dental hard tissues, commencing at about the **18th week**

!Dentine formation precedes enamel formation!



Tooth Development

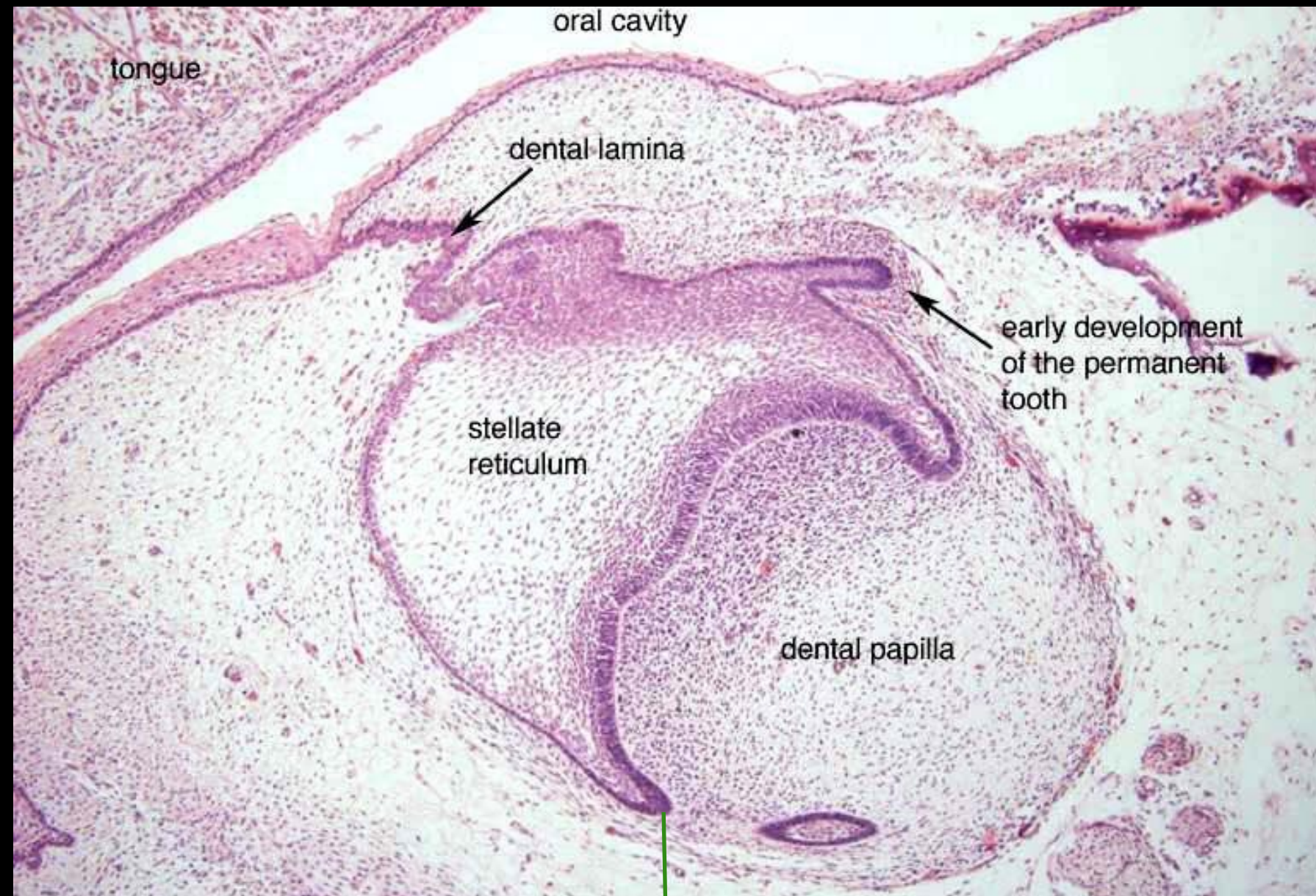
Enamel organ shows 4 distinct layers:

External enamel epithelium

Stellate reticulum

Stratum intermedium

Internal enamel epithelium



Hertwig's root sheath

Formation of the root of the tooth begins when the dental epithelial layers penetrate into the underlying mesenchyme & form the epithelial root sheath

Ectomesenchyme gives rise to the periodontium (tooth supporting tissues) —> Cement, Gingiva, Periodontal ligament, Alveolar bone

Majority of the periodontal tissues develop with formation of the roots of the teeth & tooth eruption

Majority of the periodontal tissue have an origin from dental follicle which is derived from neural crest

